



Hajali Bayramov

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ABOUT ME

Software and AI Engineer with over 8 years of experience delivering innovative solutions, currently pursuing a PhD at TU/e. Passionate about advancing interdisciplinary AI research, with expertise in natural language processing, data-driven prognostics, and AI applications for power grids. Recognized for a solution-driven mindset, strong work ethic, adaptability, and ambition to tackle complex challenges. Dedicated to enhancing the reliability, explainability, and societal impact of AI technologies.

EDUCATION AND TRAINING

10/2020 – 04/2023 Turin, Italy

MASTER OF SCIENCE Polytechnic University of Turin

- Focus on data-driven processes.
- Methodologies and technologies for data acquisition, storage, analysis, and visualization.
- Predictive and non-predictive models based on machine learning algorithms.
- Mathematical and probabilistic-statistical models for data representation, transformation, and modelling.
- Stochastic optimization and data analysis ethics and protection.
- Deeper understanding of topics, such as advanced machine learning techniques, big data analytics, or specialized domains like healthcare informatics or financial data analysis.
- More interdisciplinary courses like Innovation Management and Value Extraction from Data; a focus on understanding and managing innovation processes and emphasis on extracting value from data through strategic analysis and decision-making.

Field of study Data Science and Engineering | **Final grade** 106/110 |

Thesis Application of Artificial Intelligence for Data-Driven Prognostic of Finding Remaining Useful Life in Filters

09/2015 – 06/2020 Baku, Azerbaijan

BACHELOR OF ENGINEERING Baku Higher Oil School

- Basic engineering courses: physics, mathematics, programming, differential equations, computer architecture.
- Specialised engineering courses: electrical engineering, robotics, microprocessors, digital signal and data processing, etc.
- Management courses: economic principles, management theory, and automation engineering technical documents.

Field of study Process Automation Engineering | **Final grade** 93.28/100 |

Thesis Intelligent Monitoring System for Anomaly and Fault Detection in Complex Systems

WORK EXPERIENCE

16/05/2023 – 04/01/2024 Turin, Italy

AI/SOFTWARE ENGINEER REPLY S.P.A.

- Software Engineer for Toyota Motors Europe. The project covers the implementation of Smart Charging Stations of Electric Vehicles (keywords: Java, SpringBoot, GIT, Github Actions, StreamNative, Docker, MongoDB, Pulsar, Agile)
- Dashboard implementations
- Project on "AI implementation for developers and PM on the cloud".
- Master thesis student from 09/2022 till 02/2023, conducting research on AI pipeline for the management and predictive maintenance of a chemical mixing system.

10/2021 – 05/2022 Turin, Italy

DATA SCIENTIST/SOFTWARE ENGINEER PIAZZA DEI MESTIERI

- Experienced with programming languages such as Python, Java, C# etc.,
- Able to create SQL and Python programming modules for custom insights required by our Product Teams. Leverage statistical analysis to understand what is acceptable versus outliers.
- Able to successfully collaborate with both business users and engineers for effective analytical solution development and deployment.
- Organized data from multiple analytic platforms.
- Managed spreadsheets and performed reports.

05/2022 – 06/2022 Turin, Italy

MACHINE LEARNING ENGINEER/RESEARCHER INTERN ATELIER RIFORMA

- Researching and applying AI techniques for waste sorting and sustainable apparel up-cycling.

02/2020 – 06/2020 Baku, Azerbaijan

DATA SCIENTIST INTERN CASPIAN INNOVATION CENTER, SOCAR&IBM JV

- Conduct research to collect and assemble data for the database.
- Maintain the data integrity during extraction, manipulation, processing, analysis and storage.
- Manage database design and maintenance, administration.

DIGITAL SKILLS

Python | Java | Git | Microsoft Office | SQL | C++ | C | Linux | MongoDB | JSON | Python for data analysis (NumPy, Pandas, ...) and application development (PyQt) | Deep Learning (Tensorflow, Pytorch(basic), Jax/Flax(basic)) | Machine Learning | LaTeX (very good)

PROJECTS AND PUBLICATIONS

Projects

- Google Students data analysis
- Analysis of hotel data using python
- Intelligent Monitoring System for Anomaly and Fault Detection in Complex Systems
- Applications of Krylov methods, PCA, and SVD in real-world problems
- Applications of networks and game theory methods techniques in solving real-world problems
- Applications of Mathematics in Machine Learning in solving a real-world problem
- Applications of numerical optimization techniques (constrained and unconstrained) in solving real-world problems

Publications

- Are Local Features All You Need for Cross-Domain Visual Place Recognition? <https://arxiv.org/abs/2304.05887>
- Application of Artificial Intelligence for Data-Driven Prognostic of Finding Remaining Useful Life in Filters <https://webthesis.biblio.polito.it/26983/>

LANGUAGES

Native - Azerbaijan

- English - Advanced (C1)
Cambridge CAE
29/11/2023
C2 CEFR level, # Verification: C5036491
- Turkish - Advanced (C1)
- Italian - Upper Intermediate (B2)
PLIDA
12/2021
B1 CEFR level, # Verification: 119182

Foreign

REFERENCES

Prof. Gancho Vachkov - B.Eng thesis supervisor

gancho.vachkov@gmail.com

Prof. Danilo Giordano - M.Sc thesis supervisor

danilo.giordano@polito.it

Eng. Giovanni Federici - ICT Department Head of Piazza dei Mestieri

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